# IGCSE 0607 Practice Questions — E2.4 — Combined Questions & Markscheme

## Indices II

**Total Questions:** 6 — **Total Marks:** 35 Marks

---

## Table of Contents & Quick Navigation


- **E2.4** Indices II
  → **Paper 2 (Non-Calculator)** *(6 Qs)*


---

**1** **(a)** Simplify.

$p^4 \times p^7$

..... **[1]**

**(b)** Simplify.

$(3q^3)^2$

..... **[2]**

**(c)** Work out the value of $27^{\frac{2}{3}}$.

..... **[2]**

**2** **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

**(b)** Solve the equation.

$$2^{x+3} = 8^x$$

$x =$ ..... **[3]**

**3** **(a)** Simplify fully.

$$\frac{4x^{-\frac{1}{2}} \times 3x^2}{6x^{\frac{1}{3}}}$$

..... **[3]**

**(b)** Solve the equation.

$$9^{2x-1} \times 3^{x+1} = 27^x$$

$x =$ ..... **[4]**

**4** **(a)** Simplify.

$w^4 \times w^7$

..... **[1]**

**(b)** Simplify.

$(3x^3)^2$

..... **[2]**

**(c)** Simplify.

$$\frac{15y^8}{5y^{-2}}$$

..... **[2]**

**5** **(a)** Simplify fully.

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

**(b)** Solve the equation.

$$3^{2x+1} = 27$$

$x =$ ..... **[2]**

**(c)** Simplify fully.

$$\left(\frac{8a^6}{b^3}\right)^{\frac{1}{3}}$$

..... **[2]**

**6** **(a)** Solve the equation.

$$16^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

**(b)** Simplify fully.

$$\left(\frac{81x^{12}}{16y^{8}}\right)^{-\frac{3}{4}}$$

..... **[3]**

**1** **(a)** $p^{11}$ **[B1]**

**(b)** $9q^6$ **[B2]**

Partial correct answer $9q^k$ or $cq^6$ **[B1]**

**(c)** $(\sqrt[3]{27})^2$ or $3^2$ or $\sqrt[3]{729}$ **[M1]**

9 **[A1]**

[Back to Question] *[Total Marks: 5]*

**2** **(a)** 8 **[B1]**

$x^9$ **[B1]**

$y^6$ as denominator or $y^{-6}$ **[B1]**

**(b)** $8 = 2^3$ seen or implied by $8^x = 2^{3x}$ **[M1]**

$x + 3 = 3x$ **[M1]**

1.5 or $\frac{3}{2}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**3** **(a)** $12x^{\frac{3}{2}}$ seen as numerator **[M1]**

$2x^k$ for any $k$ or $cx^{\frac{7}{6}}$ for any $c$ **[M1]**

$2x^{\frac{7}{6}}$ **[A1]**

**(b)** Converting all terms to base 3: $3^{2(2x-1)} \times 3^{x+1} = 3^{3x}$ **[M1]**

Applying exponent multiplication law: $3^{4x-2+x+1}$ or $3^{5x-1}$ **[M1]**

Equating indices: $5x - 1 = 3x$ **[M1]**

0.5 or $\frac{1}{2}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**4** **(a)** $w^{11}$ **[B1]**

**(b)** $3^2$ or 9 seen, or $x^6$ seen **[M1]**

$9x^6$ **[A1]**

**(c)** $\frac{15}{5} = 3$ or $y^{8-(-2)}$ **[M1]**

$3y^{10}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

**5** **(a)** $4 \times \frac{3}{2} = 6$ or $x^{-3+\frac{1}{2}}$ **[M1]**

$6x^{-\frac{5}{2}}$ or $\frac{6}{x^{\frac{5}{2}}}$ **[A1]**

**(b)** $3^{2x+1} = 3^3$ or $2x + 1 = 3$ **[M1]**

1 **[A1]**

**(c)** $8^{\frac{1}{3}} = 2$ or $(a^6)^{\frac{1}{3}} = a^2$ or $(b^3)^{\frac{1}{3}} = b$ **[M1]**

$\frac{2a^2}{b}$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**6** **(a)** Expressing both sides with base 2: $(2^4)^{2x-1} = (2^3)^{x+2}$ **[M1]**

Equating powers: $4(2x - 1) = 3(x + 2)$ or $8x - 4 = 3x + 6$ **[M1]**

2 **[A1]**

**(b)** Taking the reciprocal to remove negative index: $\left(\frac{16y^8}{81x^{12}}\right)^{\frac{3}{4}}$ **[M1]**

Evaluating the fractional power of the numbers: $16^{\frac{3}{4}} = 8$ and $81^{\frac{3}{4}} = 27$ **[M1]**

$\frac{8y^6}{27x^9}$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*